

UNIVERSITI KEBANGSAAN MALAYSIA
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Wells, Wires and Dots, and What They Do

KKKC6353 Nanoelectronic Devices — Lecture 2, Document 2 of 3

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How to use this document

Document 1 ended with one big idea: trap an electron in a small enough box and it can only have certain energies, set by the size of the box. This document puts that idea to work. It shows the three kinds of trap, quantum wells, wires and dots, and the devices they make possible:

- lasers whose colour is set by the thickness of a layer;
- a laser in which one electron gives out many photons;
- the fast transistors in satellite dishes and phone chargers;
- carbon nanotube transistors;
- the quantum dots in television screens and medical imaging.

As in Document 1, each section opens with **Key ideas**, closes with **check questions**, and keeps its mathematics in optional green **Going further** boxes. Document 3 describes how these structures are made.

1 Trapping in one, two or three directions

> **Key ideas**
>
> A quantum well traps electrons in one direction, a wire in two and a dot in all three.
>
> We name each structure by the number of directions in which the electron is still free: a well is 2D, a wire 1D and a dot 0D.
>
> Traps are made by surrounding one material with another that electrons would rather avoid.

1.1 Well, wire and dot

In Document 1 the box trapped the electron in one direction only. Real structures can trap it in one, two or three directions:

- A **quantum well** is a thin layer, a few nanometres thick, sandwiched between other materials. The electron is trapped across the layer but free to move anywhere within it, like a puck on an ice rink.
- A **quantum wire** is a thin thread. The electron is trapped in two directions and free only along its length, like a train on a track.
- A **quantum dot** is a tiny crystal. The electron is trapped in all three directions and cannot move freely at all, like a bird in a small cage.

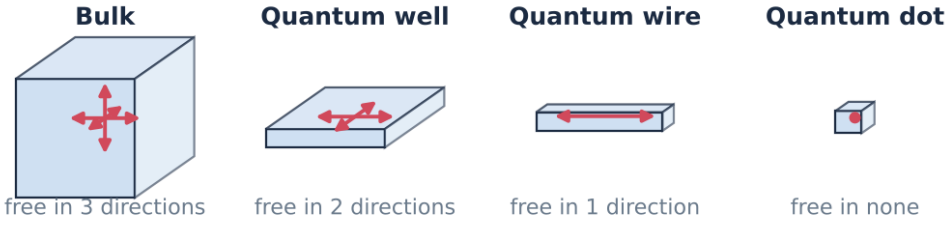


Figure 2.1. *The four cases. In bulk material the electron is free in all three directions; each extra direction of trapping removes one direction of free motion, until in a dot none is left. Figure made for this course.*

In every direction that is trapped, only certain energies are allowed, exactly as for the box of Document 1. In every direction that is free, the electron can have any energy it likes.

> **A common misconception**
>
> "A quantum dot is one-dimensional." No: a dot is **zero-dimensional**. The label counts the directions in which the electron is still *free*, not the number in which it is trapped. A well is 2D, a wire 1D and a dot 0D.

1.2 Making the walls

The walls are made from materials. Section 7 of Document 1 introduced the idea: a thin layer of GaAs between layers of AlGaAs traps electrons, because an electron has a lower energy in GaAs than in AlGaAs. Such a sandwich of different semiconductors is called a **heterostructure**. The two materials must have crystal lattices of almost the same spacing, so that the layers fit together without defects; GaAs and AlGaAs are an almost perfect fit.

Wires and dots are made by cutting a well into strips or islands, by growing tiny crystals directly, or by chemistry in a flask. Document 3 describes how.

Check your understanding

1. In how many directions is an electron free in a quantum wire?
2. Why is a quantum dot called zero-dimensional?

2 How the states are shared out

> **Key ideas**
>
> The density of states says how many places are available to electrons at each energy.
>
> Each extra direction of trapping squeezes those places into narrower ranges of energy.
>
> In a dot, all the places sit at a few exact energies, which is ideal for a laser.

2.1 Places to sit

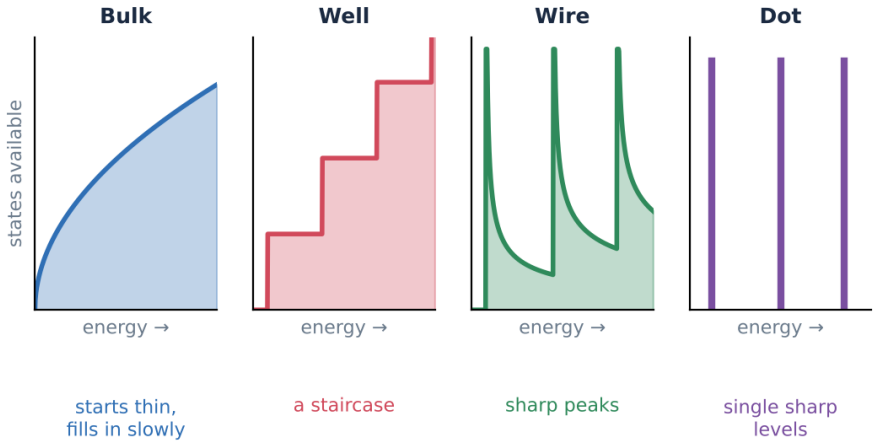


Figure 2.2. *How many places are available at each energy, for bulk material, a well, a wire and a dot. Each extra direction of trapping concentrates the places into narrower ranges of energy. Figure made for this course.*

Think of the electrons in a material as an audience filling a theatre. The seats are arranged on many levels, one for each energy, and the **density of states** is the number of seats on each level. Electrons fill the lowest levels first. How the seats are shared out across the levels depends on how the electron is trapped.

- In **bulk** material there are few places at the very bottom, and the number grows smoothly as the energy rises.
- In a **well** the places come in a staircase. Each trapped level of Document 1 opens a new step, and a full step of places is available right from its bottom edge.
- In a **wire** the places bunch into sharp peaks just above each trapped level.
- In a **dot** there are no free directions at all, so the places sit only at a few exact energies, like the levels of an atom.

2.2 Why it matters

A laser works best when all its electrons have the same energy, so that they all give out light of the same colour. In bulk material the injected electrons spread over a wide range of energies, and many of them are wasted. A well puts a whole step of places right at the bottom, so far less current is needed before the laser starts. A dot goes further, with every electron at one energy. This is why almost every semiconductor laser today uses quantum wells, and why quantum dot lasers are the next step.

Explore. The animation *Density of states* (faizverse.github.io/nanoelectronic-devices/L2) lets you switch between the four cases and watch where the electrons sit.

> **Going further: the four shapes (optional)**
> The density of states $g(E)$ is the number of electron states per unit energy. Counting the allowed waves in each case gives:
> Bulk: $g(E) \propto \sqrt{E}$, a smooth curve.
> Well: $g(E) = m^*/\pi\hbar^2$ per unit area for each subband, a constant, which gives the steps.
> Wire: $g(E) \propto 1/\sqrt{E - E_n}$ for each subband, which gives the sharp peaks.
> Dot: a single sharp line at each allowed energy E_n.

Check your understanding

1. In the theatre picture, what do the seats on each level stand for?
2. Why does a quantum well laser need less current than a bulk laser?

3 The quantum-well laser

> **Key ideas**
> A quantum-well laser has a thin well at its heart, where electrons and holes meet and give out light.
> The colour of the light is set by the thickness of the well, not only by the material.
> Nearly every semiconductor laser in use today is a quantum-well laser.

3.1 Colour from thickness

A semiconductor laser is a p–n junction. Electrons flow in from the n side and **holes**, the empty places left where electrons are missing, flow in from the p side. In a quantum-well laser they both fall into a thin well in the middle, a few nanometres thick, where they meet. When an electron drops into a hole it gives out its energy as light.

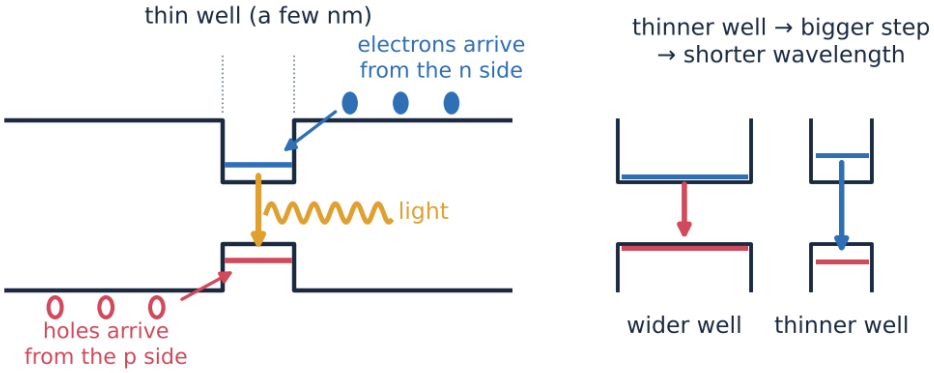


Figure 2.3. *Left: in a quantum-well laser, electrons and holes are trapped in a thin well, where they meet and give out light. Right: a thinner well pushes both trapped levels further apart, so the light has more energy and a shorter wavelength. Figure made for this course; drawn schematically.*

Because the well is a box, the electron and the hole each sit on the lowest trapped level, slightly above the bottom of their wells. The light therefore carries a little more energy than the material's band gap. Make the well thinner and those levels rise (Document 1, Section 6), so the light has more energy and a shorter wavelength. The engineer can **tune the colour by choosing the thickness**, without changing the material.

Quantum wells do more than tune the colour. As Section 2 explained, the staircase of places in a well puts many electrons at exactly the right energy, so the laser starts at a much lower current. Quantum-well lasers are in fibre-optic links, Blu-ray players, laser printers, barcode scanners and the lidar of driverless cars.

3.2 The blue LED

The white LED lamps in your home also rest on quantum wells. Their heart is a blue LED, made from very thin layers of indium gallium nitride (InGaN) between layers of gallium nitride (GaN). Making it work took decades, and Isamu Akasaki, Hiroshi Amano and Shuji Nakamura shared the 2014 Nobel Prize in Physics for it. A coating on the chip turns part of the blue light into yellow, and blue plus yellow looks white.

Explore. The animation *Quantum-well emission* lets you change the thickness of a well and watch the colour of its light change.

> **Going further: the numbers (optional)**
> The photon energy is the band gap plus the two trapped energies: $E_{\text{photon}} = E_g + E_{1e} + E_{1h}$, where E_{1e} and E_{1h} are the lowest levels of the electron and the hole in their wells. Bulk GaAs emits at 871 nm. A 10 nm GaAs well between $Al_{0.3}Ga_{0.7}As$ layers emits at about 848 nm, and a 5 nm well at about 816 nm. All three are in the near-infrared, invisible to the eye, which is the wavelength range used in many optical links and sensors.

Check your understanding

1. Where in a quantum-well laser do the electrons and holes meet?
2. A designer wants a slightly shorter wavelength from the same materials. Should the well be made thicker or thinner?

4 The quantum cascade laser

> **Key ideas**
> In an ordinary laser an electron makes one big drop, meets a hole, and is gone.
> In a quantum cascade laser an electron runs down a staircase of quantum wells and gives out a photon at every step.
> The colour is set entirely by the layer thicknesses, which opens up the infrared.

4.1 A staircase instead of a cliff

In an ordinary laser an electron falls down a "cliff", from the conduction band to the valence band, meets a hole and recombines. It gives out one photon and plays no further part. One electron, one photon.

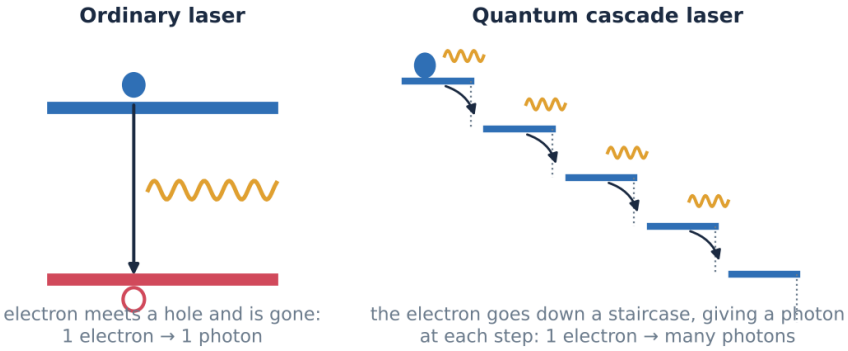


Figure 2.4. *Left: an ordinary laser. The electron makes one big drop, meets a hole and is gone. Right: a quantum cascade laser. The electron runs down a staircase of trapped levels and gives out a photon at each step. Figure made for this course; drawn schematically.*

A **quantum cascade laser** (QCL), first made by Jérôme Faist, Federico Capasso and colleagues at Bell Labs in 1994, works quite differently. It is a stack of hundreds of very thin layers that form many quantum wells in a row, and a voltage tilts the whole stack into a staircase. An electron enters at the top. At each step it drops from one trapped level to a lower one and gives out a photon. It is not destroyed: it **tunnels**, as a wave can, through a

thin wall into the top of the next step, and drops again. Like water running down a series of small waterfalls, one electron can give out a photon at every step.

Three things follow:

1. **Only electrons are needed.** No holes take part, so the QCL is called *unipolar*.
2. **The colour is set by the layers.** The size of each step depends on the thickness of the wells, not on the band gap of the material. The same materials can be designed to give any wavelength over a wide range.
3. **It reaches the infrared.** QCLs give light with wavelengths from a few to tens of micrometres, where ordinary laser diodes work poorly. Many gas molecules absorb strongly there, so QCLs can sniff out traces of a gas, a few molecules in a billion, for pollution monitoring, medical breath tests and security.

> **A common misconception**
>
> "Because one electron makes many photons, QCLs are extremely powerful." The cascade does give more light for each electron. But each photon carries only a small amount of energy, and keeping the staircase working needs a large current, so QCLs get hot and turn less of their electrical power into light than ordinary laser diodes. Their real strength is the wavelengths they reach.

Explore. The animation *Quantum cascade laser* lets you raise the voltage and watch the steps line up so that electrons can run down them.

Check your understanding

1. Why is a quantum cascade laser called unipolar?
2. What sets the colour of a QCL's light?

5 Two stories from flat and narrow channels

> **Key ideas**
>
> Electrons in a quantum well can be kept apart from the atoms that supply them, so they move unusually fast: this is the HEMT.
>
> A wire too narrow for electrons to bump into anything still has a resistance.
>
> That resistance comes in fixed steps, one for each lane the wire offers.

5.1 The HEMT: an express lane for electrons

Electrons in a semiconductor come from *donor* atoms added to the crystal. But the donors left behind are charged, and they knock passing electrons off course, slowing them down. In 1978 researchers at Bell Labs found a way round this. They put the donors in the AlGaAs layer only, next to a GaAs layer. The electrons fall into the GaAs, where their energy is lower, and gather in a thin sheet at the boundary, a quantum well. The charged donors stay behind in the other layer. The electrons now travel in an "express lane", away from the atoms that would slow them.

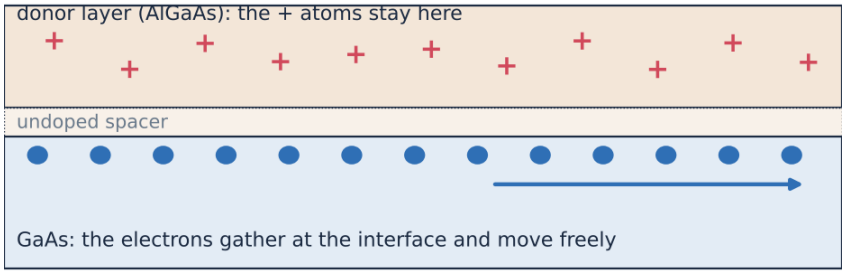


Figure 2.5. *Electrons from the donor layer fall into the GaAs and gather at the boundary, where they move freely, away from the charged donor atoms they left behind. Figure made for this course; drawn schematically.*

In 1980 Takashi Mimura and colleagues at Fujitsu put a gate on such a layer and made the **high-electron-mobility transistor**, or HEMT. HEMTs are the low-noise amplifiers in satellite dishes, radio telescopes and mobile phone base stations. A version made from gallium nitride (GaN) now handles high power too, in radar and in the small, cool-running laptop and phone chargers on sale today.

5.2 A perfect wire still has a resistance

In an ordinary wire, electrons bump into atoms and impurities all the time, and those collisions are what make resistance. A longer wire means more collisions and more resistance. Now make a wire so short and so clean that an electron crosses it without bumping into anything. This is called **ballistic** transport, as a bullet flies. You might expect zero resistance. In fact the resistance does not vanish.

Think of a motorway that narrows to a few lanes at a bridge. However smooth the bridge, traffic is limited by the number of lanes. A narrow wire offers electrons only a few "lanes", one for each trapped level across the wire, and the wide contacts at its ends must funnel electrons into them. Each lane can carry only a fixed amount of current for a given voltage. So the conductance, the ease with which current flows, comes in **fixed steps**: one step for each lane that is open.

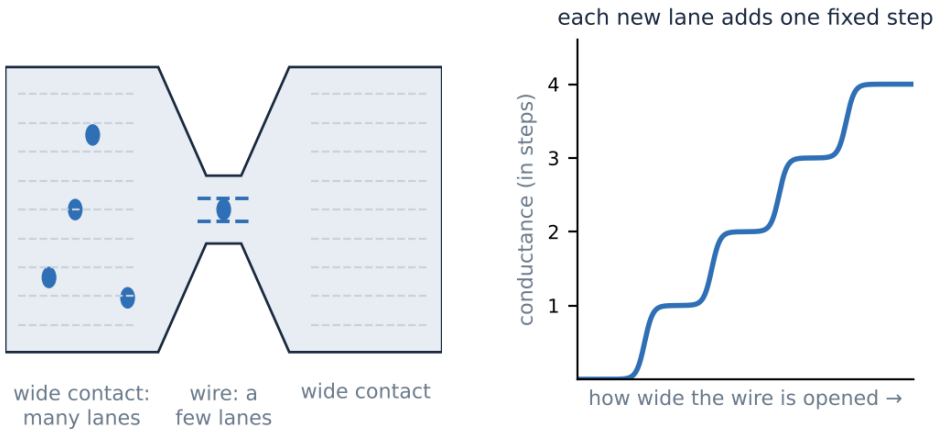


Figure 2.6. *Left: a wide contact offers many lanes; a narrow wire offers only a few. Right: as the wire is opened wider, the conductance rises in equal steps, one for each new lane. Figure made for this course; modelled on the 1988 experiments.*

This was seen in 1988 by groups in Delft and Cambridge, who squeezed a sheet of electrons into a narrow gap with gate electrodes. As they opened the gap, the conductance rose in equal steps. The resistance of each lane is about 12.9 kΩ, and it sits at the contacts, not inside the wire. No improvement to the wire can remove it.

> **Going further: the conductance quantum (optional)**
>
> Each open lane, or *mode*, contributes a conductance $G_0 = 2e^2/h = 77.5$ μS, where the 2 counts the two directions of the electron's spin. A wire with N open lanes has $G = NG_0$, so its resistance is $1/(NG_0) = 12.9\ \text{k}\Omega/N$.
>
> This result is due to Rolf Landauer (1957). It also sets a floor under the resistance of the most advanced silicon transistors, whose channels are now short enough to be nearly ballistic (Lecture 1).

Check your understanding

1. How does the HEMT keep its electrons away from the donor atoms?
2. Where does the resistance of a perfect, ballistic wire come from?

6 Carbon nanotubes and nanotube transistors

> **Key ideas**
>
> A carbon nanotube is a sheet of carbon atoms rolled into a tube about a nanometre across: a natural quantum wire.
>
> How the sheet is rolled decides whether the tube is a metal or a semiconductor.
>
> Semiconducting tubes make excellent transistors, but making billions of them reliably is still hard.

6.1 Rolling up graphene

Graphene is a single sheet of carbon atoms arranged in hexagons, like chicken wire. Roll it into a seamless tube and you have a **carbon nanotube**, about a nanometre across and up to centimetres long. Sumio Iijima identified nanotubes in 1991. With a diameter of a nanometre, a nanotube is a nearly perfect quantum wire.

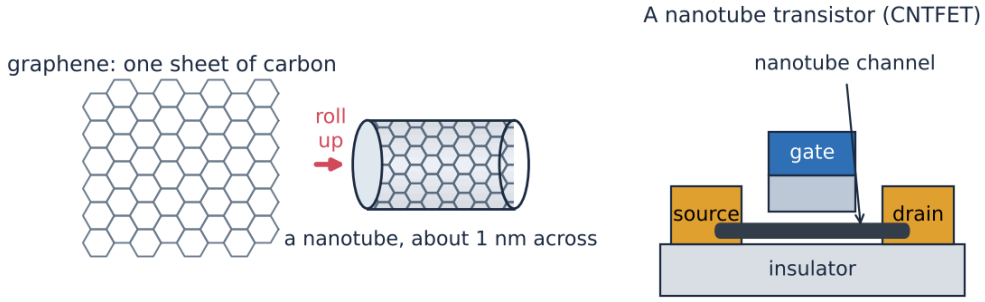


Figure 2.7. Left: rolling a sheet of graphene into a carbon nanotube. Right: a nanotube transistor, with the tube as the channel between source and drain and a gate to switch it. Figure made for this course; drawn schematically.

The sheet can be rolled straight or at an angle, like rolling a sheet of paper square or slightly askew. That choice decides how the electron waves fit around the tube, and so whether the tube conducts like a metal or behaves as a semiconductor. Statistically, about **one tube in**

three is metallic and two in three are semiconducting. For the semiconducting tubes, the thinner the tube, the larger its band gap, another case of "smaller box, bigger energy".

6.2 What makes nanotubes special

- **Electrons travel far without bumping.** In a short tube, electrons cross without scattering: transport is ballistic.
- **They move fast.** Electrons in semiconducting tubes move much more easily than in silicon.
- **They carry huge currents.** For its size, a nanotube can carry about a thousand times the current density of a copper wire.

> **A common misconception**
>
> "Ballistic nanotubes have no resistance and make no heat." Not quite. As Section 5 showed, even a perfect wire has the resistance of its lanes: about 6.5 kΩ for a metallic nanotube, which has two lanes. The heat is then made at the contacts rather than along the tube. What ballistic transport removes is the extra resistance that grows with length.

6.3 Nanotube transistors

In a **carbon nanotube field-effect transistor**, or CNTFET, one or many semiconducting tubes replace the silicon channel between source and drain, with a gate to switch them on and off. The tube is so thin that the gate controls it very well, the very property that Lecture 1 showed is hardest to keep as silicon transistors shrink.

Year	Milestone
1998	The first nanotube transistors, made from single tubes (Delft and IBM)
2013	The first nanotube computer, with 178 transistors (Stanford)
2019	A 16-bit microprocessor of 14,702 nanotube transistors (MIT)
2024	A nanotube chip of about 3,000 transistors for neural networks (Peking University)

Why are they not in your phone yet? Every growth method gives a mix of metallic and semiconducting tubes, and a single metallic tube can short-circuit a transistor. Making circuits also needs billions of tubes, all lined up in the right places, with good contacts. Researchers have made great progress on each of these problems. Lecture 4 returns to nanotube transistors alongside the other candidates to follow silicon.

Explore. The animation *Carbon nanotubes* lets you choose how the sheet is rolled and see whether the tube comes out metallic or semiconducting.

> **Going further: labelling a nanotube (optional)**
>
> How the sheet is rolled is described by a pair of whole numbers (n, m). The diameter is $d = (a/\pi)\sqrt{n^2 + nm + m^2}$, with $a = 0.246$ nm.
>
> The tube is metallic when $n - m$ is divisible by 3; otherwise it is a semiconductor with a band gap of roughly $E_g \approx 0.8$ eV nm$/d$. A 1.4 nm tube has a gap of about 0.6 eV.

Check your understanding

1. What decides whether a nanotube is metallic or semiconducting?

2. Give one reason nanotube transistors are not yet in mass production.

7 Quantum dots

> **Key ideas**
>
> A quantum dot traps electrons in all three directions, so it has only sharp, atom-like energy levels: it is an “artificial atom”.
>
> When light lifts an electron up in a dot, the electron and the hole it leaves pair up as an exciton, then recombine and give out light of one pure colour.
>
> The colour is set by the size of the dot and by its material. Dots are used in television screens and to light up cells.

7.1 Artificial atoms

A quantum dot is the box of Document 1 in all three directions. It has no free directions at all, only a set of sharp energy levels, like an atom. Unlike an atom, though, its levels can be chosen by choosing its size. That is why dots are often called **artificial atoms**.

There are two main kinds, whose making is described in Document 3:

- **Epitaxial dots**, tiny islands of one semiconductor buried inside another crystal, used in lasers.
- **Colloidal dots**, crystals 2 to 10 nm across grown in a liquid, used in displays and imaging. Moungi Bawendi, Louis Brus and Alexei Ekimov shared the 2023 Nobel Prize in Chemistry for discovering and developing them.

7.2 How a dot gives out light

When a dot absorbs a photon of ultraviolet or blue light, an electron is lifted to a higher level, leaving behind a hole. The electron and the hole attract each other and form a pair called an **exciton**. After a short time the electron drops back into the hole, and the pair’s energy comes out as a new photon.

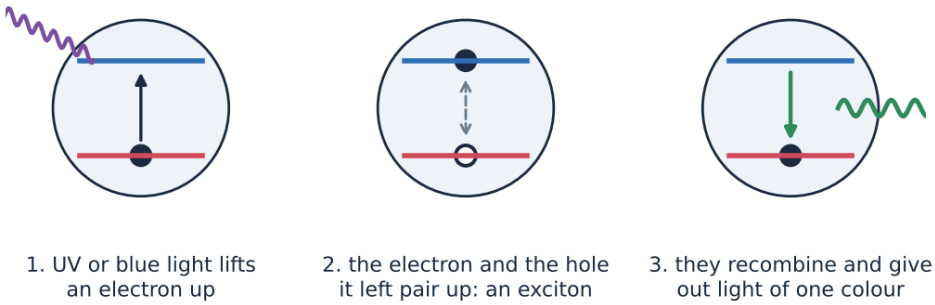


Figure 2.8. *How a quantum dot glows. Light lifts an electron up, leaving a hole; the electron and hole pair up as an exciton; then they recombine and give out light of one colour. Figure made for this course; drawn schematically.*

The colour of that light is set by the gap between the levels, and that gap is set by the size of the dot. A **small dot** traps strongly, so the gap is large and the light is **blue**. A **large dot** traps weakly, so the gap is smaller and the light is **red**. This is the row of glowing vials from the start of Document 1.

7.3 Size and material

Size tunes the colour, but only within a range set by the material. Each material has its own band gap, and no dot can give out light below it.

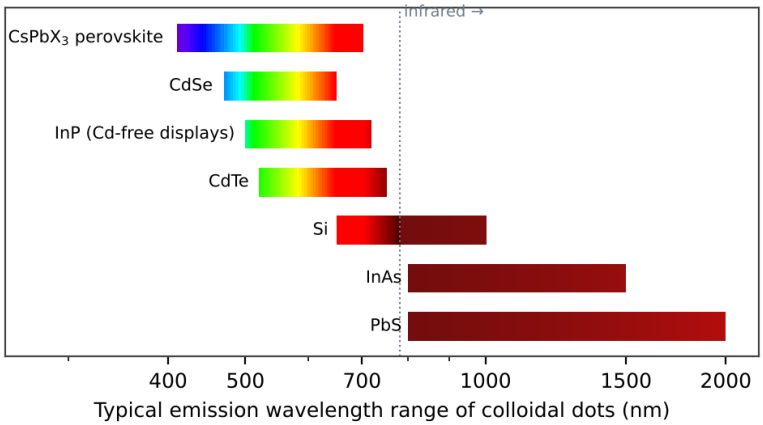


Figure 2.9. *The colours that colloidal dots of different materials can give out. CdSe and InP cover the visible range and are used in displays; PbS and InAs dots give out infrared light. Figure made for this course; ranges are approximate.*

> **A common misconception**
>
> "A small dot always glows blue and a large one red." Only within one material. A 3 nm dot of lead sulfide (PbS) glows in the infrared, while a 3 nm dot of cadmium selenide (CdSe) glows green. Size sets the colour *within* the range that the material allows.

7.4 Dots at work

Television screens. A display makes its colours by mixing red, green and blue. The purer those three are, the more colours the screen can show. Quantum dots give out very pure colours, in a narrow band set by their size. Since 2013, many LCD screens, often sold as "QLED", have used a film of red and green dots lit by blue LEDs; since 2022, "QD-OLED" screens have used dots on every pixel. Screen makers moved from cadmium-based dots to safer **indium phosphide** (InP) dots because of rules limiting cadmium in electronics.

Lighting up cells. Dots are bright and fade far more slowly than the dyes biologists use. Coated and attached to molecules that seek out a particular target, such as a cancer cell, they light it up under a microscope, and dots of several sizes can tag several targets at once in different colours. Concern about toxic elements such as cadmium means they are used widely in the laboratory but not yet routinely inside patients.

Lasers. Lasers made with dots instead of wells keep working well as they warm up, and they tolerate defects. That makes it possible to grow them directly on silicon, a route to putting lasers on silicon chips.

Explore. The animation *Quantum-dot colour* lets you change the size and material of a dot and see the colour it gives out.

> **Going further: the Brus equation (optional)**
>
> Louis Brus (1984) estimated the energy of the light from a spherical dot of radius R as
>
> $$E(R) = E_g + \frac{\hbar^2\pi^2}{2R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.8\,e^2}{4\pi\varepsilon_0\varepsilon_r R}.$$

> The first term is the band gap of the material. The second is the trapping energy of the electron and the hole, the particle in a box again, and it grows as $1/R^2$. The third is the pull between the electron and the hole, which lowers the energy a little.
>
> The equation gets the trend right but overestimates the energy of small dots. In practice, dot sizes are read from measured charts of colour against size.

Check your understanding

1. Why is a quantum dot called an artificial atom?
2. What is an exciton?
3. Why do quantum-dot screens show more vivid colours?

8 Summary, and what comes next

1. **Wells, wires and dots** trap electrons in one, two and three directions. They are named by the directions left free: 2D, 1D and 0D.
2. **Trapping shares out the available places differently**: a smooth curve in bulk, a staircase in a well, peaks in a wire and sharp levels in a dot.
3. **Quantum-well lasers** take their colour from the thickness of the well and need less current than bulk lasers.
4. **Quantum cascade lasers** send one electron down a staircase of wells, giving a photon at every step, and reach the infrared.
5. **The HEMT** keeps electrons apart from their donors so they move fast; a **perfect wire still has a resistance**, in fixed steps, one for each lane.
6. **Carbon nanotubes** are natural quantum wires; how they are rolled decides whether they are metals or semiconductors.
7. **Quantum dots** are artificial atoms whose colour is set by size and material, used in screens, imaging and lasers.

What comes next. Document 3 describes how wells, wires, dots and nanotubes are made. Lecture 3 takes up **tunnelling**, the wave leaking through a wall that we met in Document 1 and that carries electrons down the cascade laser's staircase.

Words to know

Word	Plain meaning
Heterostructure	A sandwich of different semiconductors
Quantum well, wire, dot	Traps electrons in one, two or three directions
Density of states	How many places are available to electrons at each energy
Hole	An empty place left where an electron is missing; it behaves like a positive charge
Band gap	The energy an electron must gain to start conducting; it sets the colour of light a material gives out
Ballistic transport	Electrons crossing a conductor without bumping into anything
Exciton	An electron and a hole bound together as a pair

Word	Plain meaning
HEMT	A high-electron-mobility transistor, with its electrons kept apart from the donors
CNTFET	A transistor with a carbon nanotube as its channel

Sources and further reading

Gentle introductions

- The Royal Swedish Academy of Sciences, *Quantum dots: seeds of nanoscience*, scientific background to the Nobel Prize in Chemistry 2023 (nobelprize.org).
- The Royal Swedish Academy of Sciences, *Blue LEDs: filling the world with new light*, popular science background to the Nobel Prize in Physics 2014 (nobelprize.org).

Textbooks, for the Going further boxes

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Figures. All figures were made for KKKC6353 (scripts in `figures-source/`).